Labor Input Responses to Ownership Changes: Evidence from U.S. Nursing Homes*

Joseph White[†] John Bowblis, PhD[‡]

April 6, 2026

Abstract

Ownership changes are common in U.S. nursing homes, yet their implications for care production remain unclear. Using administrative panel data from the Centers for Medicare and Medicaid Services (CMS) from 2017 to 2024, we study how ownership changes affect nurse staffing, the core input to nursing home care production. We identify control-changing ownership changes and estimate effects using a staggered-adoption difference-in-differences event-study design. We find that ownership changes lead to persistent declines in staffing, concentrated in registered nurses, certified nursing assistants, and total nursing hours per resident day, with little adjustment in licensed practical nurse staffing. Importantly, effects are substantially larger among facilities that are not chain-affiliated and before the COVID-19 pandemic. These patterns indicate that staffing is a central adjustment margin following ownership changes in a regulated setting with limited short-run pricing flexibility, suggesting that consolidation can affect resident welfare through internal production decisions even when facilities remain open.

Keywords: nursing homes, ownership change, staffing, difference-in-differences

JEL Codes: I11, I18, J22, L22, L33

[*] abc
[†] abc
[‡] abc

1 Introduction

Ownership changes are increasingly common in healthcare markets and have become a growing concern for policymakers. In U.S. Nursing homes, Skilled nursing facilities (SNFs) can change ownership through acquisitions, sales, and corporate reorganizations, and these changes are frequently talked about because of their possible implications to the quality of care (cite here). Between 2016 and 2021 more than 3,200 SNFs were sold in the United States (Prusynski et al., 2023), and this number will continue to increase well into the future (cite here). Nursing homes serve medically vulnerable residents who depend on staff for both oversight and daily living assistance, even small changes in staffing could translate into significant differences in care. We study nurse staffing as a central input to the nursing home care production, through which ownership changes may affect care delivery. Specifically we ask, how do ownership changes affect nurse staffing levels in U.S. nursing homes, and do effects differ in chains vs non-chains and before versus during the COVID-19 pandemic?

A large body of industrial organization literature studies mergers and acquisitions that reshape market structure by changing the set of competing providers in a market, raising concerns about market power, prices, wages, and competition. In healthcare, this work often evaluates consolidation through hospital, physician, or insurer mergers that increase or decrease concentration and can affect negotiated prices and input costs, making antitrust scrutiny central to the policy debate. The ownership changes we study are different. Rather than eliminating a competitor or combining two providers, most events in our data reflect a transfer of control over an existing nursing home: the facility remains in the same physical location, continues operating under the same certification framework, and continues serving a local resident population facing the same outside options and regulatory environment. As a result, these ownership changes do not mechanically alter the number of providers in the market or the underlying market structure. This distinction matters, because in settings where prices are largely fixed in the short run and market structure is unchanged, the primary channels through which ownership change can affect welfare are internal production decisions such as management practices, staffing, and other operational choices, rather than conventional market-power effects emphasized in classic industrial organization literature.

Nursing homes provide a particularly important setting in which to study the non-price consequences of ownership changes. Shortcomings in nursing home quality has been a long standing concern, and the policy debate has repeatedly emphasized incentives and constraints that shape providers' quality choices (Hackmann, 2019). Nursing home markets also exhibit severe frictions: residents and families place substantial weight on proximity, making facility decisions central to the care experienced by local populations (Gupta et al., 2021). These frictions imply that ownership changes can affect resident welfare through internal production decisions even when the facility remains open and continues to serve the same people facing the same regulations. Staffing is an especially natural outcome to study because it is both highly important for quality and one of the few outcomes that can adjust relatively quickly in response to changes in ownership.

Nursing labor is the core input in nursing home care production, and staffing is one of the most consequential operation choices facilities make. We distinguish among registered nurses (RNs), licensed practical nurses (LPNs) and certified nursing assistants (CNAs). RNs include registered nurses and directors of nursing and play central supervisory and clinical roles, including assessing residents' conditions, developing and coordinating care plans, overseeing medication-related processes, and supervising other staff. LPNs provide routine clinical services and medication management tasks such as monitoring vital signs, while CNAs deliver the majority of hands-on daily care, assisting with bathing, dressing, toileting, feeding, and mobility (Lin, 2014). Because these staff types differ in training, scope of practice, and cost, ownership changes may affect not only total staffing hours but also the skill mix of care delivered. For example, an owner seeking quick cost containment may reduce total hours or substitute away from higher-cost licensed nurses, while an owner focused on compliance or clinical quality may increase RN oversight or adjust supervision structures.

A large body of evidence indicates that staffing is directly linked to nursing home quality. Early work shows that fewer RN hours and fewer aide hours are associated with more deficiencies, including deficiencies related to quality of care and quality of life (Harrington et al., 2000). Importantly, however, staffing is an endogenous choice made jointly with other inputs and under a set of constraints such as regulation and budgets that complicate causal interpretation of simple staffing and quality correlations.

Our focus on staffing responses to ownership changes is also motivated by literature which shows that financial incentives and ownership structure can have an influence care in nursing homes. Private Equity (PE) ownership of nursing homes has been a rising policy concern, and PE ownership incentives can affect patient outcomes and operational decisions. Gupta et al. (2021) find large increases in short-term mortality associated with PE owned nursing homes and show that staffing composition changes following PE acquisition. They find declines in LPNs and CNAs alongside increases in RN hours, highlighting staffing as a mechanism linking ownership incentives to resident outcomes. While private equity represents one specific ownership form, these findings prove a more general point that is central to this paper: changes in ownership can shift objectives and constraints in ways that reshape internal production decisions, including both staffing levels and skill mix.

Research around how staffing responds to ownership changes is mixed and can be ambiguous. Ownership changes can alter managerial objectives, how owners trade off margins against quality, compliance, and reputation, and can also change the financial environment facing a facility. On one hand, new owners may face stronger incentives to reduce operating costs due to tighter budget targets or financing obligations, making labor cost reductions attractive in a sector where staffing represents a large share of operating expenses. Ownership changes may also introduce short-run disruptions such as turnover in administrators, changes in scheduling and staffing policies, contract renegotiations, and changes in recruitment and retention practices. These channels point toward reductions in staffing or substitution toward lower-cost inputs. On the other hand, ownership changes could increase staffing if new owners place greater weight on quality and compliance, seek to improve performance, or attempt to protect reputation and occupancy in a heavily monitored sector. Public reporting and quality oversight can heighten the incentive to invest in quality and staffing, particularly when quality is noticeable and easily measured (Park and Werner, 2011). In this framework, ownership changes can plausibly generate either quality reducing cost containment or quality improving investment, and the direction may differ by staff type.

This cost–quality tradeoff is highlighted by the fact that nursing home revenues are mostly fixed in the short run. Nursing homes are heavily reliant on public payers, and limited pricing flexibility can constrain the ability to finance staffing investments. Gupta et al. (2021) emphasize this reliance on subsidy and market frictions, and Bowblis and Brunt (2025) give direct evidence that financial resources are tightly linked to staffing: facilities with high Medicaid payer mix have lower staffing levels and staffing expenditures, consistent with the idea that inadequate reimbursement and thin margins limit the ability to hire and retain nursing staff.

More broadly, ownership changes can also affect the cost structure of facilities through channels that do not translate into patient care investment. Holmes (1996) shows that ownership changes are associated with higher capital related costs, plant costs, interest costs, and taxes, without corresponding increases in patient-care expenditures, implying that facility sales may shrink operating margins and increase pressure to cut variable costs such as staffing. Regulation can further shape how facilities adjust: in response to minimum quality standards, facilities may increase staffing but offset the cost through reductions in other inputs, and they may substitute across nurse types in ways that differ across settings (Bowblis and Ghattas, 2017). This institutional setting implies that staffing could rise, fall, or shift in composition following ownership change, making the net effect an empirical question and motivating a design that can detect these ambiguous adjustments.

This paper contributes to several related literatures. First, it adds to the growing body of work on ownership changes and provider performance by focusing on internal production decisions, staffing, rather than solely on market structure or prices. Second, it contributes to the nursing home literature by studying how a major organizational event, an ownership change, effects the central input into care, building on evidence that staffing improves quality (Lin, 2014) and that staffing shortfalls are linked to poor regulatory performance (Harrington et al., 2000). Third, it complements work on financial incentives and quality in nursing homes by explicitly connecting ownership transitions to staffing adjustments in an environment where financial performance can shape quality investment (Park and Werner, 2011) and where Medicaid reliance constrains staffing resources (Bowblis and Brunt, 2025). By examining heterogeneity by chain affiliation and by period relative to COVID-19, the paper sheds light on how organizational structure and shifting labor market conditions mediate staffing responses to ownership change.

To answer the research question, we use Centers for Medicare and Medicaid Services (CMS) date to construct a panel of nursing homes across the 48 contiguous U.S. states from January 2017 through June 2024. We measure staffing outcomes using hours per resident day (HPRD) for RNs, LPNs, CNAs, and total nursing hours. We estimate treatment effects using a staggered adoption difference-in-differences event study design. The results show persistent declines in nursing staffing following ownership changes, concentrated among RNs, CNAs, and total nursing hours, with heterogeneity by chain affiliation and the COVID-19 pandemic.

2 Data and Methods

2.1 Data Sources

We utilize three main sources of data: Nursing Home Compare (NHC) Archive data, Healthcare Cost Report Information System (HCRIS), and Payroll Based Journal (PBJ) data. The NHC Archive contains monthly facility level information on ownership, chain affiliation, certification status, and quality measures. The second is HCRIS, also known as the Medicare Cost Reports, which contains annual cost reports submitted by every Medicare certified nursing home in the United States. The cost reports contain information about costs, revenue, financial data for each reporting facility, and dates of ownership changes. Lastly, we use the PBJ data to obtain staffing information. The PBJ data contains the census and the number of hours worked by various types of NH staff on a daily basis.

2.2 Sample selection

Our sample is constructed at the facility-year-month level. We create our sample using the following criteria. First, we limit our sample to the 48 contiguous U.S. states from the period January 2017 to June 2024. Secondly, the NHC Archive data is matched to the HCRIS data on a unique nursing home provider number. Because HCRIS only contains information on freestanding nursing homes, our sample excludes hospital-based nursing homes. Hospital based nursing homes differ from freestanding facilities in their structure and financing. Because they are within hospitals, staffing policies and resource allocation may be jointly determined with hospital operations and respond to hospital shocks and regulations rather than nursing home specific incentives. We therefore exclude hospital based facilities to maintain a more homogeneous set of providers and to ensure ownership transitions reflect changes in nursing home management rather than broader hospital system changes. Hospital based nursing homes make up about 12 percent of all nursing homes in the United States.

2.3 Identifying Ownership Changes

No single data source provides a comprehensive record of ownership changes for U.S. nursing homes, so we combine information from HCRIS and the NHC Archive in a two-step identification and verification process to identify ownership changes.

The first step was to use the HCRIS data which contains an indicator for an ownership change, and corresponding date, in each cost report after an ownership change. We use this as our flag for ownership changes during the study period. However, these indicators may capture administrative or legal changes that do not reflect meaningful transfers of control. Thus, to verify these changes, our second step was to validate each potential ownership change in the HRCIS data with the NHC Archive data. The NHC Archive data reports ownership shares for each owner associated with a facility on a monthly basis. One issue with the NHC Archive data is that it reports the direct owners and indirect owners of each nursing home. For example, a limited liability corporation may directly own the nursing home, but that limited liability corporation is owned by different individuals. Therefore, we identified ownership changes to occur among direct and indirect owners using the following criteria:

Using this information, we define a change in ownership when a majority of ownership shares turn over between month $t-1$ and month t. Intuitively, this captures transactions in which control of the facility changes hands, including cases where a new operator acquires most or all shares. Secondly, when ownership percentages are partially or completely missing, we approximate shares by normalizing any reported percentages and, where necessary, assigning equal weights across remaining owners. To reduce false positives from purely legal restructuring, we do not flag a change in ownership if at least 80 percent of the ownership stake remains within the same surname family (e.g., a change from "Smith Holdings LLC" to "Smith Family Partners LP").

Using these verified ownership changes, the analytic sample included Nursing Homes that either (i) have no ownership change in either source or (ii) have exactly one ownership change in each source with the ownership change date within six months of one another. For these Nursing Homes, we define the change in ownership month as the first month in which the NHC Archive data indicate a change. All dates are converted to calendar months, so changes occurring at the beginning or end of a month are treated identically.

2.4 Nursing Staff Levels

Our main dependent variables are measures of nursing staff levels. The nursing staff types we use in our study are registered nurses (RNs), licensed practical nurses (LPNs), and certified nursing assistants (CNAs). RNs are the most highly trained nursing staff and are responsible for assessment, care planning, medication administration, and supervision of other nursing personnel. LPNs provide basic medical care under the supervision of RNs and physicians, including monitoring vital signs and administering certain treatments. CNAs provide the majority of direct daily care to residents, assisting with activities of daily living. Because these staff types differ in training, scope of practice, and compensation, changes in staffing composition may reflect different organizational responses to ownership changes. We therefore examine each staff category separately, as well as total nursing hours per resident day.

Nursing staff levels were calculated from PBJ data, which reports the number of hours worked and the census for each type of nursing staff on a daily basis. We used this information to calculate nursing staff levels in hours per resident day (HPRD) as follows: for each facility i, calendar month t, and staff type, we first aggregate daily hours to the monthly level and then compute

$$\text{HPRD}_{i,t} = \frac{H_{i,t}}{R_{i,t}}, \tag{1}$$

where $H_{i,t}$ is the total hours worked by the staff type during month t and $R_{i,t}$ is the total resident days in that facility-month. Total HPRD is defined as the sum of RN, LPN, and CNA HPRD.

We construct two additional measures of data quality. First, we define a monthly coverage ratio

$$C_{i,t} = \frac{\text{DaysReported}_{i,t}}{\text{DaysInMonth}_t},$$

where $\text{DaysReported}_{i,t}$ is the number of distinct days in month t with complete PBJ records for facility i. This measure allows us to identify incomplete data in our sample and describe our data completeness. We use this measure for sample construction and descriptive purposes, excluding facilities and observations with low reporting coverage.

Second, we track gaps in the reporting for each facility. For each observation we compute *gap*, defined as the number of calendar months since the previous PBJ observation for that facility. This captures discontinuities in reporting that may arise from administrative faults or temporary periods of not reporting PBJ system, for example during the COVID-19 period. Our baseline specification places no restriction on reporting gaps, however; we use the gap measure in robustness analyses to assess whether our results are sensitive to excluding observations following reporting gaps.

Finally, we exclude a small number of observations with implausible staffing levels: months in which RN and LPN HPRD are both zero, total HPRD is below 1.5 or above 12, or CNA HPRD exceeds 5.25. These observations are outliers that likely reflect reporting errors rather than genuine variation in staffing. These exclusion criteria align with the criteria outlined in the CMS NHC Technical Users Guide.

2.5 Other Controls

To account for other factors that are correlated with nursing staff levels, we included a number of control variables that are consistent with other studies (cite here). These control variables were measured at the facility-level and came from NHC and HCRIS. The following variables are used as controls in all specifications: Ownership type (government, non-profit, and for-profit), chain affiliation, total number of beds, occupancy rate, payer mix (percent of revenue coming from Medicare, and percent of revenue coming from Medicaid), and acuity. Our measure of acuity is case-mix provided by CMS in the NHC archive data. During our study period CMS altered the methodology used to construct case-mix indices. As a result, raw case-mix values are not directly comparable over time. To address this issue, we measure acuity using relative rankings rather than levels. Specifically, we construct case-mix quartiles within each state and calendar month.

2.6 Analytic Approach

Ownership changes occur at different dates across facilities in our study period, so identification relies on staggered adoption within a two-way fixed effects event-study framework.

Recent literature shows that conventional two-way fixed effects (TWFE) estimators can produce biased estimates in staggered adoption settings when treatment effects are heterogeneous over time (Goodman-Bacon, 2021). In particular, TWFE estimates can place negative weights on certain comparisons and implicitly rely on already treated units as controls. To circumvent this, we employ a staggered adoption event study regression design. This approach allows us to estimate treatment effects before and after ownership changes while assessing pre-trends. Specifically we estimate the following event study regression:

$$Y_{it} = \sum_{\tau=-24}^{24} \beta_\tau \, \mathbf{1}\{\text{event\_time}_{it} = \tau\} \cdot \text{Treat}_i + X_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}. \tag{2}$$

where Y_{it} denotes nursing staff levels in HPRD for facility i in in month t. The variable event_time_{it} measures the number of months relative to the month of ownership change. Thus, the coefficients β_τ capture the evolution of the nursing staff levels before and after ownership transitions. The vector X_{it} includes time-varying facility characteristics, while α_i and λ_t denote facility and month fixed effects, respectively. Identification comes from within-facility changes in staffing relative to the timing of ownership transitions, using facilities that have not yet experienced a change as controls.

In addition to estimating using levels, we also estimate specifications where the dependent variable is expressed in logarithms, $\log(Y_{it})$. Log specifications allow coefficients to be interpreted as approximate percentage changes in staffing and reduce sensitivity to differences in baseline staffing levels across facilities. For outcomes with zero staffing in a given month, log specifications are omitted.

2.6.1 Parallel Trends

The identifying assumption underlying our model design is that, absent an ownership change, staffing outcomes in treated facilities would have followed the same trends as those in not-yet treated and untreated facilities. We assess the plausibility of this assumption by examining estimated coefficients on the pre-treatment event-time indicators and by conducting joint

Wald tests of whether the pre-treatment coefficients are jointly equal to zero.

A key complication in this setting is that the recorded ownership-change date may not coincide with the date at which the ownership transition begins to affect facility operations. Ownership changes often occur in stages. First, a transaction is negotiated and a contract is signed. Then, there is typically an interim period before the formal closing or legal transfer of control is completed and recorded in the data. During this interim period, managers and prospective owners may already begin making adjustments, including changes to staffing. As a result, staffing may begin to change before the recorded ownership-change month.

This creates a problem of anticipation in the pre-treatment window. The administrative date lags the underlying organizational change meaning the parallel trends assumption might not hold. If staffing is already adjusting in the months just before the recorded event month, then those observations are not a clean untreated baseline. This contaminates the immediate pre-period and can lead standard pre-trend tests to reject parallel trends even when the longer-run counterfactual trends are otherwise plausible.

Table **??** reports p-values for individual pre-treatment event-time coefficients across outcomes, separately for specifications that include all observations (Panel A) and for specifications that exclude the anticipatory period immediately preceding ownership changes (Panel B). Under the parallel trends assumption, we would expect pre-treatment coefficients to be statistically indistinguishable from zero, implying relatively large p-values and no systematic pattern of significance across pre-treatment periods.

Table 1: Joint Wald Tests of Pre-trends (Event Study)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: Levels (HPPD)				
2 Year Full Pre-Window	107.49 (23)	41.63 (23)	46.13 (23)	51.22 (23)
	[0.0000]	[0.0100]	[0.0029]	[0.0006]
2 Year Window with Donut	34.83 (20)	36.57 (20)	17.83 (20)	24.92 (20)
	[0.0210]	[0.0132]	[0.5984]	[0.2046]
1 Year Window with Donut	7.55 (8)	10.49 (8)	3.89 (8)	6.08 (8)
	[0.4788]	[0.2322]	[0.8666]	[0.6387]
Panel B: Logs (HPPD)				
2 Year Full Pre-Window	78.21 (23)	45.72 (23)	34.54 (23)	49.22 (23)
	[0.0000]	[0.0032]	[0.0578]	[0.0012]
2 Year Window with Donut	17.14 (20)	37.42 (20)	13.28 (20)	22.26 (20)
	[0.6436]	[0.0104]	[0.8651]	[0.3268]
1 Year Window with Donut	3.33 (8)	13.48 (8)	7.50 (8)	6.33 (8)
	[0.9122]	[0.0965]	[0.4837]	[0.6102]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.
Tested windows and reference periods: 2 Year Full Pre-Window tests $\tau = -24$ to $\tau = -2$ with reference $\tau = -1$; 2 Year Window with Donut tests $\tau = -24$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$); 1 Year Window with Donut tests $\tau = -12$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$).
Sample sizes (rows): 2 Year Full Pre-Window ($N = 632,231$); 2 Year Window with Donut ($N = 628,107$); 1 Year Window with Donut ($N = 628,107$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.

Table 1 reports the results of the joint Wald tests. In the "2 Year Full Pre-Window" specification, which includes the full set of pre-treatment months ($\tau = -24$ to $\tau = -2$) and uses $\tau = -1$ as the reference period, the joint tests reject the null of no pre-trends for several outcomes, particularly for RN and total staffing in both levels and logs. This indicates that staffing begins to move in the periods leading up to the recorded ownership transition when the months immediately preceding the change are included in the pre-window, consistent with anticipatory adjustment around the transition date.

We therefore report "donut" specifications that exclude the three months immediately preceding ownership changes ($\tau \in \{-3, -2, -1\}$) and use $\tau = -4$ as the reference period. The donut specification is intended to remove the portion of the pre-period most likely to be contaminated by anticipation and timing mismatch between the formal administrative transfer date and the onset of operational change. Relative to the full pre-window tests, the donut specifications yield substantially weaker evidence of pre-trend violations for several outcomes, with joint tests that are no longer statistically distinguishable from zero in the shorter one-year donut window. Overall, these results suggest that the months immediately before the recorded change month are not a valid untreated baseline, and that excluding $\tau \in \{-3, -2, -1\}$ improves the plausibility of the parallel trends assumption.

Accordingly, our preferred specifications estimate equation (2) excluding observations in $\tau \in \{-3, -2, -1\}$ and interpret dynamic treatment effects relative to $\tau = -4$. Standard errors are clustered at the facility level to account for serial correlation within nursing homes.

While our main specification relies on the event-study design, we also present TWFE estimates to summarize the average post ownership change effect. We estimate the following regression:

$$Y_{it} = \beta_{\text{post}}\, Post_{it} + X_{it}\boldsymbol{\gamma} + \alpha_i + \lambda_t + \varepsilon_{it}, \tag{3}$$

where Y_{it} denotes nurse staffing outcomes for staff types RN, LPN, CNA, or total HPRD for facility i in month t. The indicator $Post_{it}$ equals one for all months following an ownership change and zero otherwise. Facility fixed effects α_i absorb time invariant differences across nursing homes, while month fixed effects λ_t control for common shocks to staffing over time.

The coefficient β_{post} captures the average change in staffing following an ownership transition, relative to the pre-change period. All specifications include the same time varying covariates as in the event-study models, and standard errors are clustered at the facility level.

2.6.2 Subgroup Analysis

We further examine heterogeneity in the effects of ownership change with two further analyses: the COVID-19 pandemic and baseline chain affiliation.

First, we estimate separate models for the pre-pandemic period (January 2017 through December 2019) and the pandemic period (April 2020 through June 2024). Labor market conditions for nursing staff changed substantially during the COVID-19 pandemic, including heightened turnover, changes in wages, and staffing shortages. Estimating models separately by period allows us to assess whether ownership changes had differential effects on staffing before and during this structural shift in the nursing labor market.

Second, we examine heterogeneity by baseline chain status. We classify facilities based on whether they were affiliated with a chain at the beginning of the study period (January 2017), prior to any ownership transitions in our sample. Chain affiliated facilities differ systematically from independent facilities in ways that are directly relevant for staffing decisions. Prior work shows that chains operate internal labor markets, share managerial expertise across facilities, and are better positioned to reallocate staff in response to shocks, potentially lessening the effects of ownership changes on staffing outcomes. In contrast, independent facilities may face tighter local labor constraints and fewer organizational buffers, making staffing levels more sensitive to changes in ownership and management.

By looking at baseline chain status, we compare facilities with similar organizational structures prior to treatment, rather than confounding the effects of ownership change with changes in chain affiliation occurring at the same time. Estimating models separately for chain and non-chain facilities therefore allows us to assess whether ownership transitions have different effects on staffing across organizational structures, and whether chain affiliation dampens or magnifies staffing adjustments following ownership changes.

In both subgroup analyses, we estimate the same specifications as in equations (2) and (3), maintaining consistent controls, fixed effects, and clustering. These analyses shed light on mechanisms and contextual factors that shape the staffing consequences of ownership changes.

3 Results

3.1 Summary Statistics

Table 2 reports summary statistics for the main analysis sample. The final sample consists of approximately 7,806 unique skilled nursing facilities and 632,231 facility-month observations spanning January 2017 through June 2024. Of these facilities,1589 experience an ownership change during the sample period.

Table 2: Summary Statistics

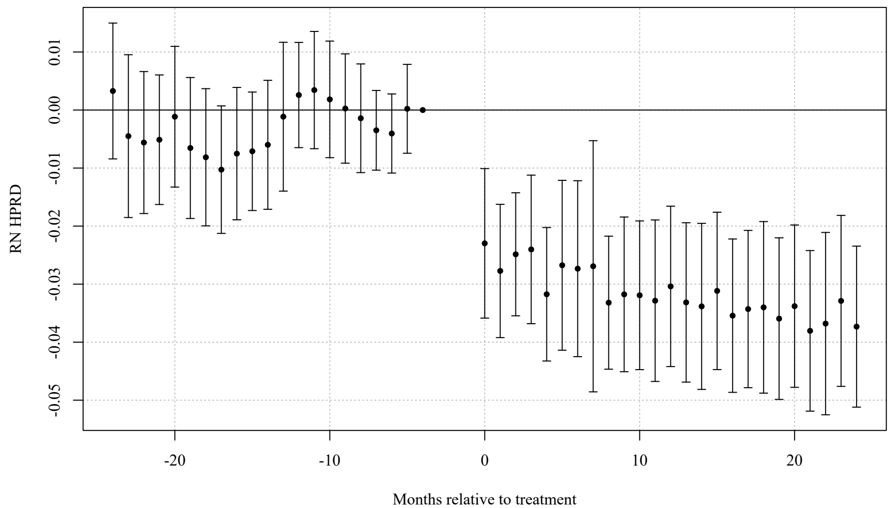
Variable	**Mean**	**SD**
Panel A: Outcome variables		
RN HPRD	0.427	0.309
LPN HPRD	0.796	0.325
CNA HPRD	2.100	0.548
Total HPRD	3.324	0.793
Panel B: Control variables		
Government (dummy)	0.101	0.301
Non-profit (dummy)	0.171	0.377
Chain affiliation (dummy)	0.546	0.498
Beds	108.4	57.9
Occupancy rate (%)	76.4	15.5
% Medicare	13.0	11.1
% Medicaid	60.5	20.2
Acuity quartile 2 (state-month)	0.231	0.422
Acuity quartile 3 (state-month)	0.230	0.421
Acuity quartile 4 (state-month)	0.233	0.423

Notes: The unit of observation is facility–month. HPRD denotes hours per resident day. RN, LPN, and CNA denote registered nurses, licensed practical nurses, and certified nursing assistants, respectively. Total HPRD is the sum of RN, LPN, and CNA HPRD. Occupancy rate is the ratio of residents to certified beds (percent). % Medicare and % Medicaid are the shares of residents covered by Medicare and Medicaid, respectively. Government and Non-profit are ownership-type indicator variables (for-profit omitted category). Chain affiliation is an indicator for membership in a multi-facility chain. Acuity quartiles are state–month quartiles of resident acuity (case-mix) with quartile 1 omitted. Rows = 632,231; Facilities = 7,806; Treated facilities = 1,589; Period = 2017/01–2024/06; Average months per facility = 81.0.

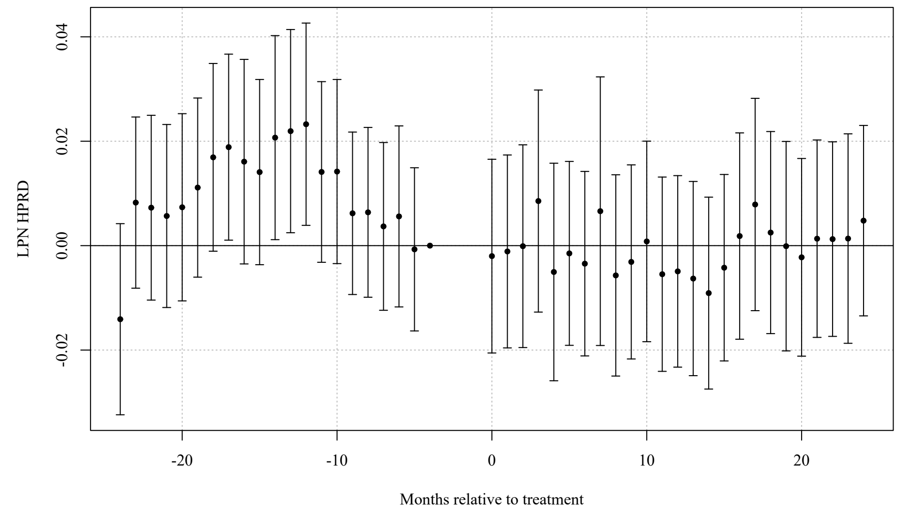
3.2 Main Results

Given our main analysis suggests that parallel trends assumption likely holds, Figure 1 presents event study estimates of the effect of ownership changes on nursing staffing outcomes (i.e., Equation 2). Following an ownership transition, staffing declines persist for many periods, with the largest and most precise effects occurring for registered nurses (RN), certified nursing assistants (CNA), and total nursing hours per resident day.

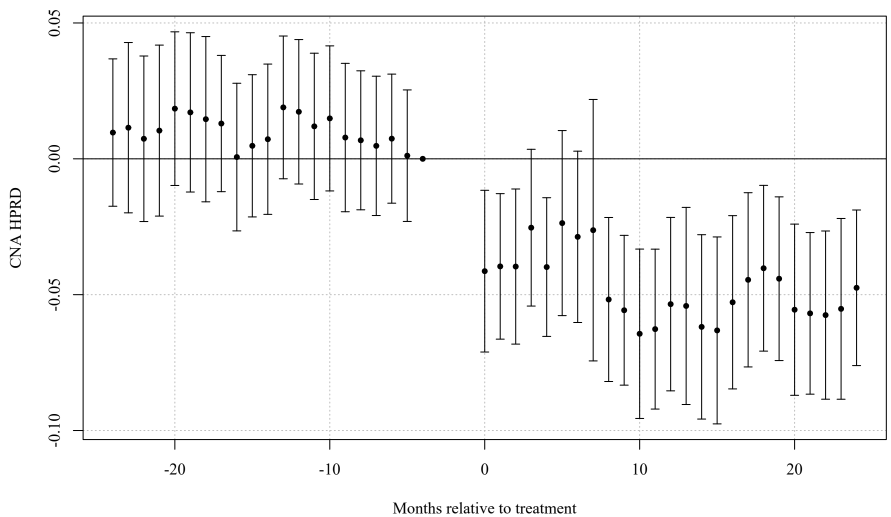
RN staffing declines immediately following ownership changes and remains far below pre-transition levels throughout the post-treatment period. CNA staffing exhibits a similar but slightly smaller decline, while total nursing hours mirror the combined reductions across staff types. In contrast, LPN staffing shows limited and statistically insignificant changes over time. These dynamic patterns indicate that ownership transitions lead to sustained reductions in staffing rather than short run adjustments.



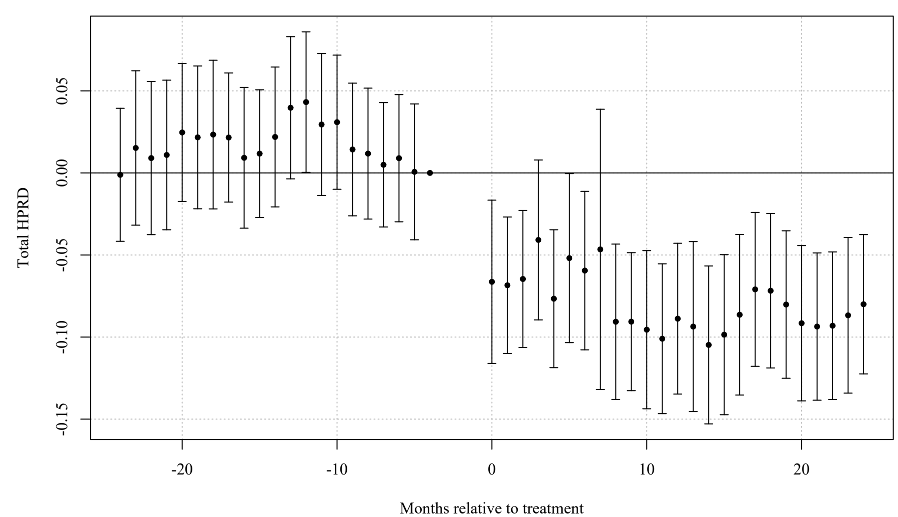
(a) Registered Nurses (RN)



(b) Licensed Practical Nurses (LPN)



(c) Certified Nursing Assistants (CNA)



(d) Total Nursing Hours

Figure 1: Dynamic Effects of Ownership Change on Nursing Staffing. Event study estimates of the effect of a change in ownership on nursing staff levels in hours per resident day (HPRD). Points show coefficient estimates relative to the reference period $\tau = -4$; vertical bars denote 95% confidence intervals. The three months prior to the ownership change ($\tau = -3, -2, -1$) are excluded. All specifications include control variables and facility and month fixed effects.

To summarize these dynamic effects with a single average treatment estimate, Table 3 reports two-way fixed effects estimates of the ownership change effect (i.e., Equation 3). Panel A reports results in levels, while Panel B reports logged results. Consistent with the event study results, ownership changes lead to meaningful reductions in staffing levels. RN staffing declines by approximately 0.03 hours per resident day (roughly 9 percent), CNA staffing falls by about 0.05 hours (approximately 3 percent), and total staffing decreases by 0.08 hours per resident day (roughly 3 percent). LPN staffing shows no statistically significant change. Estimates are nearly identical whether the observations for the three month period before the ownership change occurred is included or excluded from the model.

Table 3: Two-Way Fixed Effects Estimates of *post* on Staffing Outcomes (Baseline, Without anticipation)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
HPRD	-0.032^{***}	0.000	-0.054^{***}	-0.086^{***}
	(0.005)	(0.006)	(0.009)	(0.013)
Log(HPRD)	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
	(0.016)	(0.009)	(0.005)	(0.004)

Notes: Each cell reports the coefficient on *post* with two-way clustered standard errors (by facility and month) in parentheses. The table reports staffing levels (HPRD) and log staffing levels (Log(HPRD)). Sample: Without anticipation ($N_{\text{HPRD}} = 628,107$; $N_{\text{Log}} = 628,107$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

3.3 Effects Before and During Covid-19 Pandemic

Table 4 examines whether these effects differ before and after the COVID-19 pandemic. The magnitude of the decline is roughly halved in the post pandemic period for RN, CNA, and total HPRD, suggesting that pandemic induced shifts in labor markets and staffing substantially reduced the impact of ownership transitions.

Table 4: TWFE Estimates of *post*: Pre- vs Post-pandemic Periods (Without anticipation)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: Staffing Levels in HPRD				
Pre-Pandemic Period (2017/01 - 2019/12)	-0.029^{***}	-0.016^{*}	-0.063^{***}	-0.108^{***}
	(0.006)	(0.008)	(0.013)	(0.019)
Pandemic Period (2020/04 - 2024/06)	-0.018^{**}	-0.000	-0.041^{***}	-0.058^{***}
	(0.007)	(0.008)	(0.015)	(0.020)
Panel B: Log Staffing Levels in HPRD				
Pre-Pandemic Period (2017/01 - 2019/12)	-0.068^{***}	-0.009	-0.032^{***}	-0.033^{***}
	(0.024)	(0.012)	(0.006)	(0.006)
Pandemic Period (2020/04 - 2024/06)	-0.063^{***}	0.002	-0.020^{**}	-0.019^{***}
	(0.021)	(0.011)	(0.008)	(0.006)

Notes: Each cell reports the coefficient on *post* with two-way clustered standard errors (by facility and month) in parentheses. Panel A reports levels (HPRD); Panel B reports logs (HPRD). Sample sizes: Row 1 ($N_{\text{levels}} = 255,005$; $N_{\text{logs}} = 255,005$), Row 2 ($N_{\text{levels}} = 359,413$; $N_{\text{logs}} = 359,413$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.
Pre-pandemic 2017/01–2019/12; Pandemic 2020/04–2024/06.

3.4 Effects by Chain Status

Table 5 compares effects for facilities that began the study period as part of a chain and those that did not. Staffing reductions are noticeably larger among non chain facilities. CNA HPRD falls by 0.034 hours in chain facilities and by 0.079 hours in non chains, and total HPRD declines by 0.062 versus 0.113 hours, respectively. On a log scale, these differences translate into substantially larger percentage declines for non-chain facilities. LPN staffing again shows minimal movement.

Table 5: TWFE Estimates of *post*: Chain vs Non-chain Facilities (Jan 2017 Baseline, Without anticipation)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: Staffing Levels in HPRD				
Chain January 2017	-0.031^{***}	0.003	-0.034^{***}	-0.062^{***}
	(0.007)	(0.008)	(0.011)	(0.015)
Non-chain January 2017	-0.037^{***}	0.003	-0.079^{***}	-0.113^{***}
	(0.009)	(0.011)	(0.019)	(0.025)
Panel B: Log Staffing Levels in HPRD				
Chain January 2017	-0.099^{***}	0.008	-0.019^{***}	-0.020^{***}
	(0.021)	(0.013)	(0.006)	(0.005)
Non-chain January 2017	-0.085^{***}	0.013	-0.037^{***}	-0.032^{***}
	(0.030)	(0.017)	(0.010)	(0.008)

Notes: Each cell reports the coefficient on *post* with two-way clustered standard errors (by facility and month) in parentheses. Panel A reports levels (HPRD); Panel B reports logs (HPRD). Sample sizes: Row 1 ($N_{\text{levels}} = 322,100$; $N_{\text{logs}} = 322,100$), Row 2 ($N_{\text{levels}} = 243,036$; $N_{\text{logs}} = 243,036$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.
Baseline chain classification determined by facility status in January 2017.

Overall, ownership changes lead to sizable and persistent reductions in nursing staffing, with effects concentrated among RN, CNA, and total hours per resident day. These reductions are larger prior to the COVID-19 pandemic and among facilities that were not part of a chain at baseline. Together, the event study and TWFE results indicate that ownership transitions systematically worsen staffing conditions in nursing homes rather than inducing temporary adjustments.

3.5 Robustness

This section presents a series of robustness checks designed to assess the validity of our identifying assumptions and the sensitivity of our results to alternative model choices. Across all exercises, the estimated effects of ownership change on staffing remain stable in sign, magnitude, and statistical significance. Most of these will have tables/plots that will go in the appendix.

3.5.1 Assumptions

Our main event-study specification is estimated using two-way fixed effects (TWFE) with staggered treatment timing. In this setting, TWFE can implicitly use already-treated facilities as controls for cohorts treated later, which may distort event-study estimates when treatment effects are dynamic or heterogeneous across cohorts. To assess sensitivity to this issue, we re-estimate the event study using a stacked difference-in-differences design.

In the stacked approach, we construct cohort-specific samples for each treatment month g. For a given cohort, treated facilities are those with an ownership change in month g, and the control group consists only of facilities that are never treated or not-yet-treated relative to g (i.e., with treatment dates after g). We restrict each cohort-specific sample to a symmetric event window around g and then stack these cohort datasets into a single analysis file. Estimation includes the same covariates as the baseline specification and fixed effects that absorb facility-level time-invariant heterogeneity and common calendar shocks; standard errors are clustered at the facility level.

Appendix Figures 2 through 5 show that the stacked event-study estimates closely track the baseline TWFE event-study results: the post-transition dynamics and the relative magnitudes across staffing types remain qualitatively unchanged. In addition, Appendix Table 11 reports joint Wald tests of pre-treatment coefficients in the stacked specification (under the donut design used in the baseline), which are consistent with flat pre-trends in the remaining pre-period. Together, these results suggest that the main findings are not driven by treated-as-control comparisons that can arise in TWFE under staggered adoption.

3.5.2 Event Window and "Donut"

As we have previously discussed, because of changes to staffing in anticipation of a formal ownership change, the assumptions of parallel trends is violated. As a result of this, in our baseline specifications we "donut" equation 2 and equation 3 by removing $\tau \in \{-3, -2, -1\}$. To test the validity of removing this set of pre-treatment coefficients, we estimate equations 2 and 3 with different "donut" sets and show that the estimated post-treatment effects remain largely unchanged. Row 6 of Table 10 reports estimates of equation 3 with a wider anticipatory period excluded from the sample. The results from row 6 show no differences in values or statistical significance. Similarly, row 7 reports estimates under a smaller anticipatory window that is excluded from the data, and we see no differences in the results.

3.5.3 Gap Size

Because staffing data is occasionally missing or reported with gaps, we examine whether our results are sensitive to the length of time between consecutive observations. Restricting the sample to facilities with smaller reporting gap shows that estimates are similar to those in the full sample, indicating that data gaps do not drive our findings. Row 2 of Table 10 reports our results for equation 3 with the restriction that facilities with a reporting gap of larger than 6 are excluded from our sample. Compared to the baseline (row 1) this result is identical across staff types and statistical significance. The exact same result is true for row 3 which reports estimates for equation 3 where facilities with a reporting gap of greater than 3 are removed. Row 4 and 5 report estimates where gap greater than 1 or equal to 0 respectively, are removed. Similarly, we see no differences in these results with respect to the baseline.

4 Discussion

We find that ownership changes are followed by persistent reductions in nurse staffing, particularly in RN hours, CNA hours, and total nursing hours per resident day. In a regulated sector with limited short-run pricing flexibility, staffing is one of the most immediately adjustable margins, so changes in staffing provide a useful window into how new owners reshape internal production decisions. Our event study estimates, together with the results across alternative specifications, suggest that these are not short-lived reductions around the recorded ownership change month, but sustained adjustments following a change in control.

4.1 No declines in LPN hours

One notable result is that LPN staffing shows little to no decline following ownership change, even as RN and CNA hours fall. This pattern is consistent with substitution across nurse types arising from differences in training, scope of practice, and how tasks are organized within nursing homes. RNs perform higher-skill clinical and supervisory functions, while CNAs provide the majority of hands-on daily care (Lin, 2014). LPNs occupy an intermediate role where they can perform a range of routine clinical tasks that overlap with RN responsibilities, while also helping maintain day-to-day coverage in ways that partly offset reductions in CNA time.

This makes LPNs a plausible way to adjust when new owners seek to reduce labor costs without generating large immediate changes in service provision. CNAs generally cannot substitute for RNs on licensed or supervisory tasks, and RNs are not typically used to replace CNA-level direct-care work because of wage differences and the organization of production. LPNs, by contrast, can absorb some routine clinical work that might otherwise require RN time, while also supporting continuity of care when CNA hours are reduced. In that sense, the relative stability of LPN staffing is consistent with compositional adjustment rather than uniform staffing cuts.

4.2 Are the staffing reductions economically meaningful?

An important question is whether the estimated staffing reductions are large in economic terms. Benchmarking the estimates against baseline staffing levels helps provide context. In our sample, mean staffing levels are 0.427 RN HPRD, 0.796 LPN HPRD, 2.100 CNA HPRD, and 3.324 total HPRD. In the baseline TWFE specification, an ownership change reduces RN staffing by 0.032 HPRD, CNA staffing by 0.054 HPRD, and total staffing by 0.086 HPRD, while LPN staffing does not change significantly.

These effects correspond to declines of roughly 7.5% for RN staffing, 2.6% for CNA staffing, and 2.6% for total staffing. Expressed in more intuitive terms, 0.032 RN HPRD is about 1.9 fewer RN minutes per resident-day, 0.054 CNA HPRD is about 3.2 fewer CNA minutes per resident-day, and 0.086 total HPRD is about 5.2 fewer total nursing minutes per resident-day.

Our estimates should not be interpreted as implying large welfare effects simply because they are statistically significant. At the same time, staffing is a core input into care production, so even modest staffing changes could still matter, especially in facilities operating close to staffing constraints. In larger facilities, some reductions may be diffused across many residents and workers; in smaller facilities, comparable average reductions may be more likely to bind and produce noticeable changes in quality. Thus, the economic significance of these estimates for resident care is ultimately ambiguous in our paper. What we can show is that ownership changes affect staffing inputs; whether those input changes translate into meaningful changes in deficiencies, hospitalizations, mortality, or other outcomes remains an open empirical question.

4.3 Heterogeneity

Two heterogeneity patterns provide additional insight into mechanisms. First, staffing declines are larger for non-chain facilities than for chain-affiliated facilities. This pattern is consistent with organizational buffers and internal labor market capacity in chains: chain facilities may have greater ability to stabilize staffing, share managerial practices, and respond to financial or operational shocks without reducing measured staffing as sharply. In contrast, independent facilities may face tighter constraints and fewer margin-stabilizing tools, making staffing a more immediate adjustment channel. Second, estimated staffing reductions are smaller during the COVID-19 period than in the pre-pandemic period. A natural interpretation is that pandemic-era labor market tightness limited owners' ability to implement further staffing reductions, while heightened scrutiny and operational disruptions may have constrained the feasible set of staffing adjustments.

4.4 Implications

A key contribution of this paper is to establish staffing as an adjustment margin following ownership changes, even in a setting where prices are largely fixed and production decisions are heavily regulated. However, staffing is an intermediate input, and whether these ownership-induced staffing reductions reflect efficiency gains (e.g., reduced slack, improved task allocation) or quality-reducing cost containment is ultimately an empirical question about outcomes. A natural extension is to link ownership transitions to resident-centered measures (e.g., deficiencies, rehospitalizations, mortality, or other quality indicators) and test whether the staffing reductions documented here coincide with changes in measured quality. Such analysis could also help with the interpretation of the LPN substitution mechanism: if RN and CNA hours fall while quality is unchanged, this would be more consistent with task reallocation and efficiency; if quality deteriorates, it would indicate that staffing reductions likely reflect quality-reducing cost containment. The results here motivate a broader evaluation of ownership changes that jointly considers staffing composition and resident outcomes.

4.5 Limitations

First, despite the two-step approach to identifying ownership transitions and the use of a donut window to address anticipatory adjustments, the timing of ownership change may still be measured with error, and some operational changes may precede the recorded transition date. Second, while the event-study design with fixed effects and pre-trend diagnostics helps address confounding, ownership changes may coincide with unobserved shocks (e.g., financial distress) that independently affect staffing trajectories. Third, PBJ-based staffing measures may be affected by reporting completeness; although robustness checks suggest the findings are not driven by reporting gaps, measurement error remains possible. Finally, because the analysis focuses on staffing inputs, the paper does not directly estimate effects on resident outcomes; interpreting welfare implications therefore requires either stronger assumptions about the staffing–quality relationship or additional outcome analyses.

5 Conclusion

Ownership changes are common in U.S. nursing homes, yet their implications for care production are not well understood. Using panel data from 2017 through 2024, we identify ownership transitions of true economic control and estimate their effects on nurse hours per resident day. We find that ownership changes lead to declines in RN staffing, CNA staffing, and total nursing hours per resident day, while LPN staffing remains stable. These effects are larger in non-chain facilities and smaller during the COVID-19 period.

Our findings suggest that ownership changes affect nursing home operations through internal production decisions, even when the set of providers in a market does not change. In a sector where short-run prices are heavily constrained, staffing is an especially important adjustment margin. At the same time, the estimated reductions are modest in absolute terms, so the welfare implications for residents cannot be determined from staffing evidence alone. The results should therefore be interpreted as evidence that ownership transitions change staffing inputs, not as definitive evidence about resident outcomes.

This paper contributes to the literature by shifting away from ownership changes solely as a market-structure event and toward its consequences for the internal organization of care. A natural next step is to link ownership transitions to resident outcomes such as deficiencies, hospitalizations, and mortality, and to study the mechanisms through which new owners alter staffing and task allocation. Doing so would help clarify when ownership transitions reflect efficiency-enhancing reorganization and when they instead reflect quality-reducing cost containment.

References

Antonipillai, V., Ng, E., Baumann, A., Crea-Arsenio, M., and Garner, R. (2025). Staffing levels and expenses in canadian long-term care facilities by ownership status before and during the covid-19 pandemic. *Health Reports*, 36(7):3–14.

Bowblis, J. R. and Brunt, C. S. (2025). Nursing home staffing expenditures and levels are impaired by high medicaid payer-mix. *Journal of the American Medical Directors Association*, 26(8):105723.

Bowblis, J. R. and Ghattas, A. (2017). The impact of minimum quality standard regulations on nursing home staffing, quality, and exit decisions. *Review of Industrial Organization*, 50:43–68.

Castle, N. G. (2005). Nursing home closures, changes in ownership, and competition. *Inquiry*, 42(3):281–292.

Cohen, J. W. and Spector, W. D. (1996). The effect of medicaid reimbursement on quality of care in nursing homes. *Journal of Health Economics*, 15(1):23–48.

Dall'Ora, C., Saville, C., Rubbo, B., Turner, L., Jones, J., and Griffiths, P. (2022). Nurse staffing levels and patient outcomes: A systematic review of longitudinal studies. *International Journal of Nursing Studies*, 134:104311.

Fan, Y. (2013). Ownership consolidation and product characteristics: A study of the us daily newspaper market. *American Economic Review*, 103(5):1598–1628.

Fan, Y. and Yang, C. (2022). Estimating discrete games with many firms and many decisions: An application to merger and product variety. Working Paper 30146, National Bureau of Economic Research.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.

Grabowski, D. C. and Bowblis, J. R. (2023). Minimum-staffing rules for u.s. nursing homes — opportunities and challenges. *New England Journal of Medicine*, 389(18):1637–1640.

Grabowski, D. C. and Stevenson, D. G. (2008). Ownership conversions and nursing home performance. *Health Services Research*, 43(4):1184–1203.

Gupta, A., Howell, S. T., Yannelis, C., and Gupta, A. (2021). Owner incentives and performance in healthcare: Private equity investment in nursing homes. Working Paper 28474, National Bureau of Economic Research.

Hackmann, M. B. (2019). Incentivizing better quality of care: The role of medicaid and competition in the nursing home industry. *The American Economic Review*, 109(5):1684–1716.

Harrington, C., Zimmerman, D., Karon, S. L., Robinson, J., and Beutel, P. (2000). Nursing home staffing and its relationship to deficiencies. *The Journals of Gerontology: Series B*, 55(5):S278–S287.

Haucap, J. and Stiebale, J. (2023). Non-price effects of mergers and acquisitions. DICE Discussion Papers 402, Heinrich Heine University DÃ¼sseldorf, DÃ¼sseldorf Institute for Competition Economics (DICE).

Holmes, J. S. (1996). The effects of ownership and ownership change on nursing home industry costs. *Health Services Research*, 31(3):327–346.

Juvé-Udina, M.-E., Adamuz, J., González-Samartino, M., Tapia-Pérez, M., Jiménez-Martínez, E., Berbis-Morello, C., Polushkina-Merchanskaya, O., Zabalegui, A., and López-

Jiménez, M.-M. (2025). Association between nurse staffing coverage and patient outcomes in a context of prepandemic structural understaffing: A patient-unit-level analysis. *Journal of Nursing Management*, 2025(1):8003569.

Kelly, E., Propper, C., and Zaranko, B. (2026). Nursing shortages and patient outcomes. *Journal of Health Economics*, 105:103082.

Lichtenberg, F. R. and Siegel, D. (1990). The effect of ownership changes on the employment and wages of central office and other personnel. *The Journal of Law & Economics*, 33(2):383–408.

Lin, H. (2014). Revisiting the relationship between nurse staffing and quality of care in nursing homes: An instrumental variables approach. *Journal of Health Economics*, 37:13–24.

McGuckin, R. H. and Nguyen, S. V. (2001). The impact of ownership changes: a view from labor markets. *International Journal of Industrial Organization*, 19(5):739–762. Competition Policy in Dynamic Markets.

Park, J. and Werner, R. M. (2011). Changes in the relationship between nursing home financial performance and quality of care under public reporting. *Health Economics*, 20(7):783–801.

Prusynski, R. A., Humbert, A., and Mroz, T. M. (2023). Skilled nursing facility changes in ownership and short-stay medicare patient outcomes. *JAMA Network Open*, 6(9):e2334551–e2334551.

Sutton, N., Ma, N., Yang, J. S., and Lin, J. (2024). Quality effects of home acquisitions in residential aged care. *Australasian Journal on Ageing*, 43(1):158–166.

Zhang, X. and Grabowski, D. C. (2004). Nursing home staffing and quality under the nursing home reform act. *The Gerontologist*, 44(1):13–23.

Tables

Table 6: TWFE Post Estimates with Interaction: Chain vs Non-chain (Without anticipation)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: Staffing Levels in HPPD				
Chain (baseline 2017Q1): *post*	-0.038^{***}	-0.000	-0.054^{***}	-0.092^{***}
	(0.006)	(0.007)	(0.011)	(0.015)
Difference (Non-chain − Chain): *post* × Non-chain	0.006	0.005	-0.016	-0.005
	(0.011)	(0.013)	(0.021)	(0.028)
Non-chain effect: *post* + interaction	-0.032^{***}	0.005	-0.070^{***}	-0.097^{***}
	(0.009)	(0.011)	(0.019)	(0.025)
Panel B: Log Staffing Levels				
Chain (baseline 2017Q1): *post*	-0.127^{***}	-0.003	-0.030^{***}	-0.030^{***}
	(0.020)	(0.012)	(0.006)	(0.005)
Difference (Non-chain − Chain): *post* × Non-chain	0.054	0.014	-0.003	0.003
	(0.036)	(0.021)	(0.011)	(0.009)
Non-chain effect: *post* + interaction	-0.073^{**}	0.011	-0.033^{***}	-0.027^{***}
	(0.029)	(0.017)	(0.010)	(0.008)

Notes: Each cell reports the indicated coefficient with two-way clustered standard errors (by facility and month) in parentheses.
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Sample: facilities with non-missing baseline chain status (2017Q1), without anticipation ($N = 561,440$).
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 7: TWFE Post Estimates with Interaction: Pre-pandemic vs Pandemic (Without anticipation)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: Staffing Levels in HPPD				
Pre-pandemic: *post*	-0.023^{***}	-0.016^{**}	-0.040^{***}	-0.079^{***}
	(0.006)	(0.008)	(0.012)	(0.017)
Difference (Pandemic − Pre): *post* × Pandemic	-0.010^{*}	0.019^{***}	-0.017	-0.009
	(0.006)	(0.007)	(0.011)	(0.015)
Pandemic effect: *post* + interaction	-0.033^{***}	0.002	-0.057^{***}	-0.088^{***}
	(0.005)	(0.006)	(0.010)	(0.013)
Panel B: Log Staffing Levels				
Pre-pandemic: *post*	-0.076^{***}	-0.021^{*}	-0.014^{**}	-0.022^{***}
	(0.024)	(0.012)	(0.007)	(0.005)
Difference (Pandemic − Pre): *post* × Pandemic	-0.020	0.027^{***}	-0.016^{**}	-0.005
	(0.022)	(0.010)	(0.006)	(0.005)
Pandemic effect: *post* + interaction	-0.096^{***}	0.006	-0.030^{***}	-0.027^{***}
	(0.017)	(0.009)	(0.005)	(0.004)

Notes: Each cell reports the indicated coefficient with two-way clustered standard errors (by facility and month) in parentheses.
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Sample: 2017/01–2019/12 and 2020/04–2024/06 (excluding 2020/01–2020/03), without anticipation ($N = 610,364$).
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 8: TWFE DiD Estimates of *post* on Staffing Outcomes (Stacked Sample, Baseline Donut)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
HPPD	-0.023^{***}	-0.007	-0.051^{***}	-0.082^{***}
	(0.004)	(0.005)	(0.008)	(0.011)
Log(HPPD)	-0.066^{***}	-0.003	-0.027^{***}	-0.026^{***}
	(0.014)	(0.008)	(0.004)	(0.003)

Notes: Each cell reports the coefficient on *post* with facility-clustered standard errors in parentheses. The stacked sample includes never-treated and not-yet-treated controls for each cohort; the donut excludes $\tau \in \{-3, -2, -1\}$. Sample sizes: $N_{\text{HPPD}} = 23,877,687$; $N_{\text{Log}} = 23,492,034$.
Specifications include facility-by-cohort fixed effects (stack_id), calendar-month fixed effects, cohort fixed effects, and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 9: Two-Way Fixed Effects Estimates of *post* on Staffing Outcomes Using MCR Event Timing

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: With anticipation				
HPRD	-0.031^{***}	-0.000	-0.052^{***}	-0.083^{***}
	(0.005)	(0.006)	(0.009)	(0.012)
Log(HPRD)	-0.088^{***}	0.003	-0.026^{***}	-0.025^{***}
	(0.016)	(0.009)	(0.005)	(0.004)
Panel B: Without anticipation				
HPRD	-0.032^{***}	-0.000	-0.054^{***}	-0.086^{***}
	(0.005)	(0.006)	(0.009)	(0.013)
Log(HPRD)	-0.094^{***}	0.003	-0.028^{***}	-0.026^{***}
	(0.016)	(0.009)	(0.005)	(0.004)

Notes: Each cell reports the coefficient on *post* with two-way clustered standard errors (by facility and month) in parentheses.

Panel A uses the full baseline sample with anticipation periods included. Panel B excludes observations with *anticipation2*=1 (that is, event time $\in \{-3, -2, -1\}$).

All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators, when available in the panel.

Estimation sample sizes for Panel A HPRD models: RN=632,227; LPN=632,227; CNA=632,227; Total=632,227.

Estimation sample sizes for Panel A log(HPRD) models: RN=626,782; LPN=627,689; CNA=632,111; Total=632,227.

Estimation sample sizes for Panel B HPRD models: RN=628,115; LPN=628,115; CNA=628,115; Total=628,115.

Estimation sample sizes for Panel B log(HPRD) models: RN=622,716; LPN=623,586; CNA=627,999; Total=628,115.

Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 10: TWFE estimates of *post* (without anticipation).

	N	Outcomes RN	LPN	CNA	Total
Panel A: Staffing Levels in HPPD					
(1) Baseline	628,107	-0.032^{***}	0.000	-0.054^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
(2) Sample excludes *gap* > 6	627,923	-0.032^{***}	-0.000	-0.054^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
(3) Sample excludes *gap* > 3	627,626	-0.032^{***}	-0.000	-0.054^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
(4) Sample excludes *gap* > 1	623,585	-0.032^{***}	0.000	-0.055^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
(5) Sample excludes *gap* > 0	623,585	-0.032^{***}	0.000	-0.055^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
(6) Drop $t \in \{-4, -3, -2, -1\}$	626,729	-0.032^{***}	0.000	-0.055^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
(7) Drop $t \in \{-2, -1\}$ only	628,107	-0.032^{***}	0.000	-0.054^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
Panel B: Log Staffing Levels in HPPD					
(1) Baseline	628,107	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)
(2) Sample excludes *gap* > 6	627,923	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)
(3) Sample excludes *gap* > 3	627,626	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)
(4) Sample excludes *gap* > 1	623,585	-0.092^{***}	0.004	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)
(5) Sample excludes *gap* > 0	623,585	-0.092^{***}	0.004	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)
(6) Drop $t \in \{-4, -3, -2, -1\}$	626,729	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)
(7) Drop $t \in \{-2, -1\}$ only	628,107	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)

Notes: Each cell reports the *post* coefficient with two-way clustered standard errors at the facility and month levels. Panel A uses staffing levels (HPPD); Panel B uses logs of HPPD.
All specifications include facility and month fixed effects and controls for ownership (government, non-profit, chain), beds, occupancy rate, payer mix (Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 11: Joint Wald Tests of Pre-trends (Stacked Event Study) — Levels

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
2 Year Window with Donut	28.46 (20)	35.10 (20)	20.25 (20)	24.08 (20)
	[0.0990]	[0.0196]	[0.4423]	[0.2388]
1 Year Window with Donut	5.26 (8)	24.35 (8)	6.59 (8)	15.99 (8)
	[0.7300]	[0.0020]	[0.5817]	[0.0425]

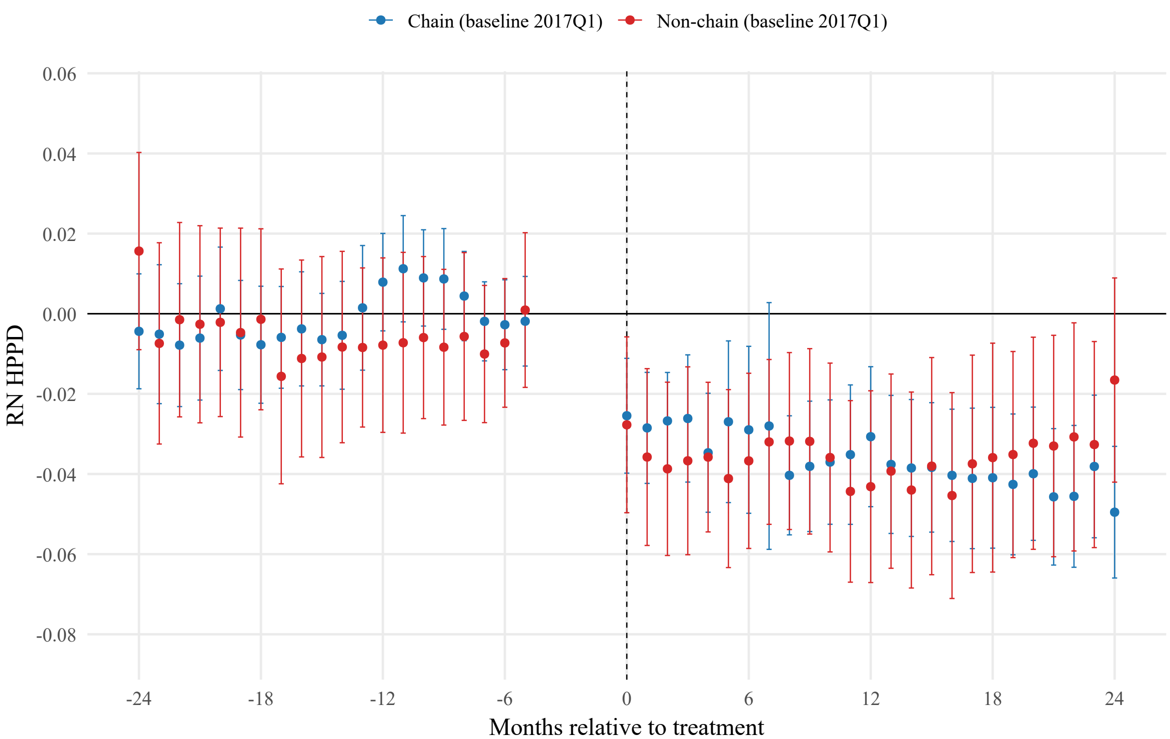
Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.
Tested windows and reference periods: 2 Year Donut tests $\tau = -24$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$); 1 Year Donut tests $\tau = -12$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$).
Sample sizes (stacked rows): 2 Year Donut ($N = 23,877,687$); 1 Year Donut ($N = 13,121,085$).
All specifications include stacked-unit fixed effects (facility-by-cohort), calendar-month fixed effects, cohort fixed effects, and baseline covariates. Standard errors are clustered at the facility level.

Table 12: Joint Wald Tests of Pre-trends Using MCR Event Timing

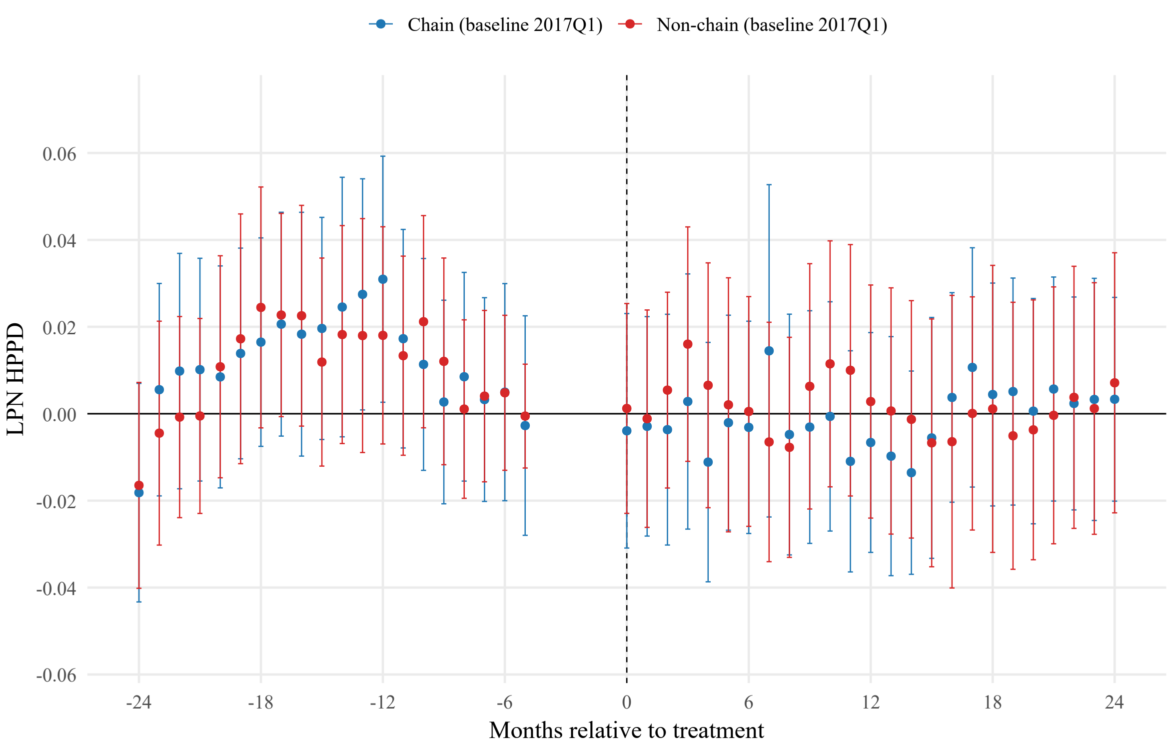
	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: Levels (HPPD)				
2 Year Full Pre-Window	94.45 (23)	39.43 (23)	35.94 (23)	48.62 (23)
	[0.0000]	[0.0178]	[0.0418]	[0.0014]
2 Year Window with Donut	27.85 (20)	34.12 (20)	11.33 (20)	23.85 (20)
	[0.1129]	[0.0253]	[0.9373]	[0.2491]
Panel B: Logs (HPPD)				
2 Year Full Pre-Window	47.13 (23)	37.17 (23)	35.68 (23)	44.39 (23)
	[0.0022]	[0.0312]	[0.0445]	[0.0047]
2 Year Window with Donut	26.19 (20)	36.25 (20)	10.89 (20)	22.75 (20)
	[0.1595]	[0.0144]	[0.9491]	[0.3012]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.
Tested windows and reference periods: 2 Year Full Pre-Window tests $\tau = -24$ to $\tau = -2$ with reference $\tau = -1$; 2 Year Window with Donut tests $\tau = -24$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$).
Sample sizes (rows): 2 Year Full Pre-Window ($N = 632,231$); 2 Year Window with Donut ($N = 628,119$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.

Figures



Chain (baseline 2017Q1)
Non-chain (baseline 2017Q1)
RN HPPD
0.06
0.04
0.02
0.00
-0.02
-0.04
-0.06
-0.08
-24
-18
-12
-6
0
6
12
18
24
Months relative to treatment



Chain (baseline 2017Q1)
Non-chain (baseline 2017Q1)
LPN HPPD
0.06
0.04
0.02
0.00
-0.02
-0.04
-0.06
-24
-18
-12
-6
0
6
12
18
24
Months relative to treatment

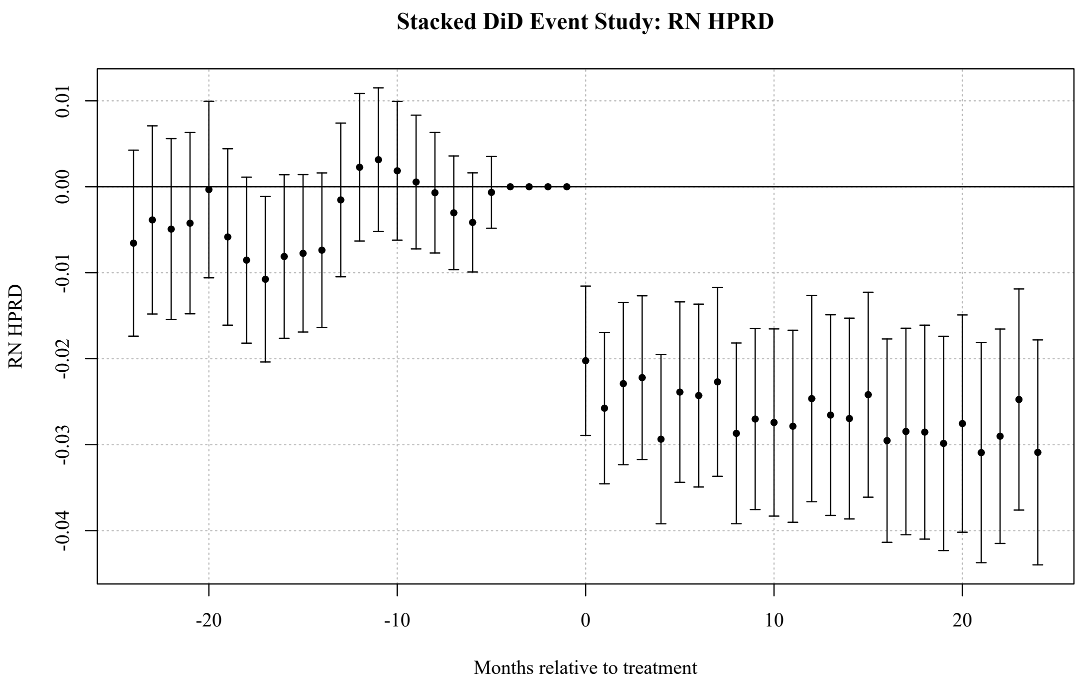


Figure 2: Stacked event-study estimates (RN staffing, levels). The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

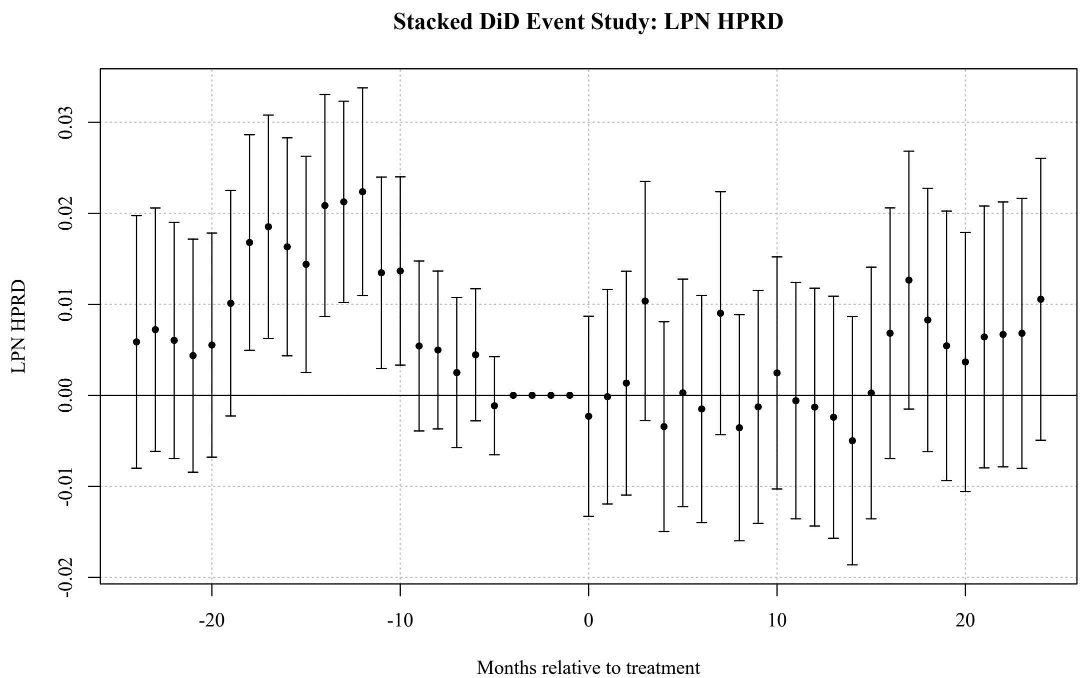


Figure 3: Stacked event-study estimates (LPN staffing, levels). The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

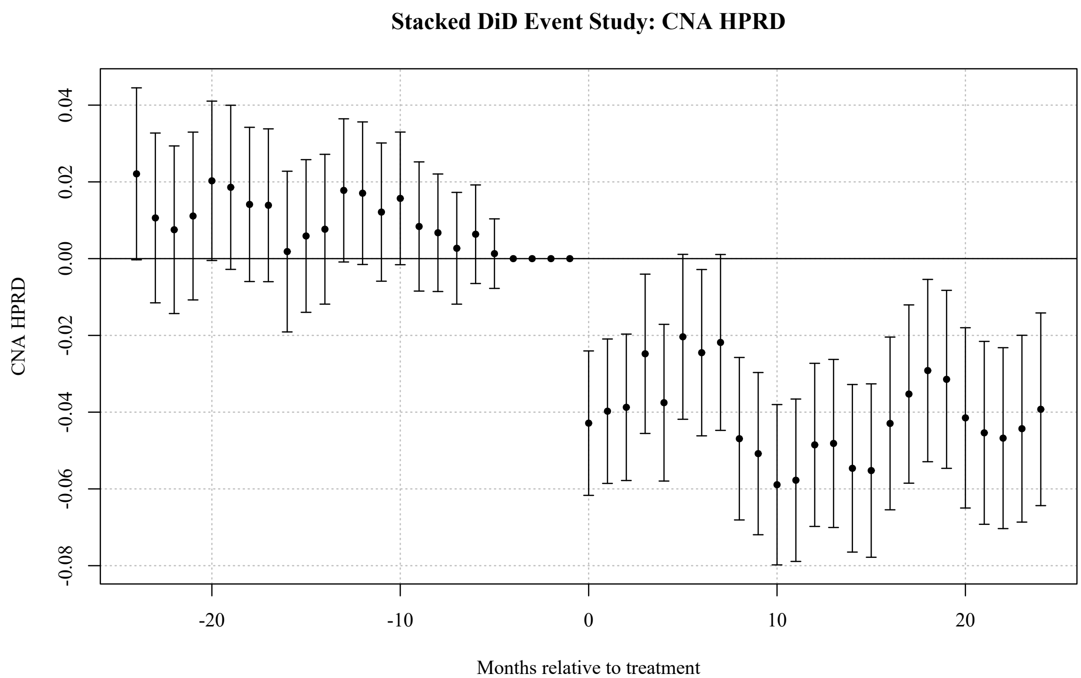


Figure 4: Stacked event-study estimates (CNA staffing, levels). The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

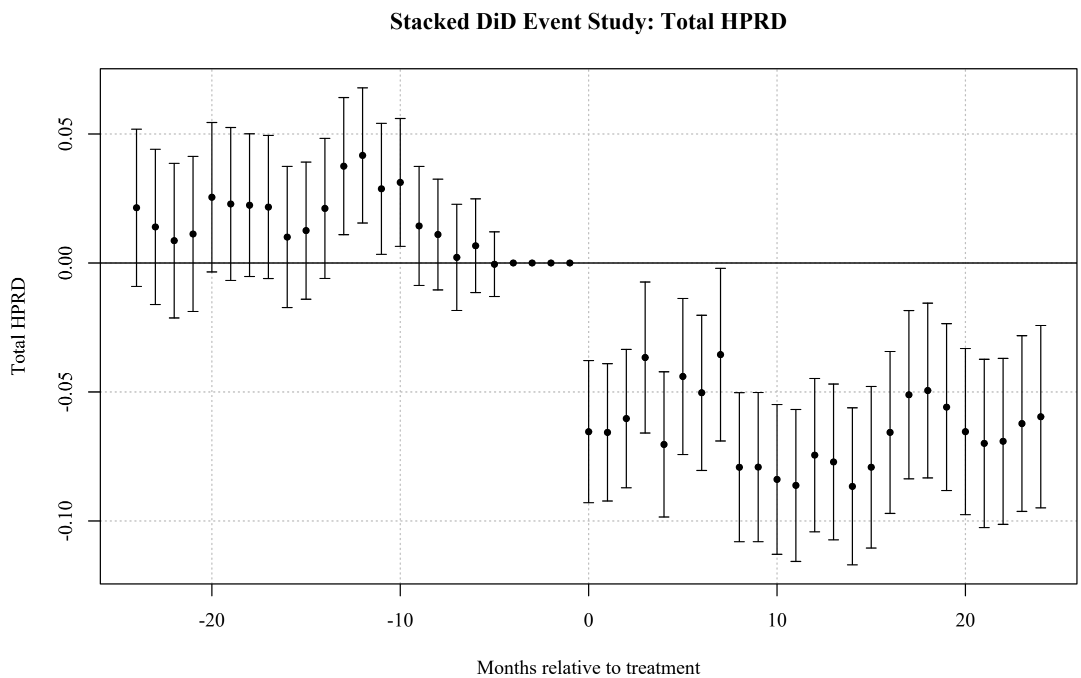


Figure 5: Stacked event-study estimates (Total staffing, levels). The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

Appendix A. Placeholder